THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVEVERNMENT

FORM TWO MOCK EXAMINATION MARKING SCHEME 2018

LONGIDO DISTRICT

BIOLOGY

1. 10 Marks @1mark

i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
B	**B**	**A**	**D**	**E**	**A**	**C**	**B**	**C**	**B**

2. 10 Marks@ 1mark
   (i) TRUE
   (ii) FALSE
   (iii) FALSE
   (iv) FALSE
   (v) FALSE
   (vi) TRUE
   (vii) TRUE
   (viii) FALSE
   (ix) TRUE
   (x) TRUE
3. 5 Marks@1mark
   (i) C
   (ii) F
   (iii) J
   (iv) A
   (v) G
4. 5 Marks@1mark
   (i) BICUSPID VALVE
   (ii) PULMONARY VEIN

(iii) VEINS

(iv) ARTERIOSCLEROSIS

(v) HYPERTENSION

5. (i) Laboratory is the special room where biology scientific experiments are conducted. 1Mark

(ii) Five qualities of a good biology laboratory are

- It must have big enough window to allow free movement of air
- It must allow enough light
- It must have fire extinguisher
- It must have adequate space for carrying out experiment, storage of specimens and apparatus
- It must have first aid kit
- It must have good supply of water

5points @1mark=5Marks

b) Functions of

(i) Mortar and pestle; It used to grind solid substances

(ii) Petri dish; It used to grow micro organisms in the laboratory or Placement of specimen for close observation

(iii) Hand lens; It used to magnify small objects so that can be easily to be observed

(iv) Thermometer; It used to measure temperature

4 Marks@ 1mark

6. (a) Gaseous exchange is the movement of oxygen and carbon dioxide across a respiratory surface.

In mammal the gaseous exchange happen as the result of Inhalation (Breathing in air into the lungs and Exhalation (Breathing out air from the lungs). **0.5 Marks**

Mechanisms

During inhalation (2.5 Marks)

- The muscle of the diaphragm contract, pulling the diaphragm downwards
- As this happens the external intercostals muscles contract and pull the ribcage upward and outwards
- The result of these movements is an increase in the volume and decrease in air pressure of the thorax
- This makes air rush into the lungs through the nostril, trachea and bronchioles.

During exhalation (2.5 Marks)

- The muscles of the diaphragm relax and the diaphragm resumes its dome shape
- The external intercostals muscles relax pulling the ribcage inward
- As the result the volume of thorax decreases and the pressure inside increases
- This force the air out through the bronchioles, trachea and nostril

(b) 4.5 Marks@ 0.5mark

ORGANISMS	RESPIRATORY SURFACES
Amoeba	CELL MEMBRANE
Insects	TRACHEAL SYSTEM
Spider	BOOK LUNG
Fish	GILLS
Plants	LEAVES,STEM,ROOTS
Amphibians	SKIN,GILLS,and LUNGS
Mammals	LUNGS
Birds	LUNGS
Reptiles	LUNGS

7. (a) (i) Food web **2marks**

(ii) Food chain **2marks**

(b) (i) Secondary consumers are LION and HYENA **2Marks**

(ii) Primary producers are GRASS and SHRUBS **2Marks**

(c) Mucor is a decomposer; so it recycles nutrients back to the soil where they can be taken by primary producers in sustaining life and producing food again **2Marks**

8. (a) (i) Tissue is the collection of cells of same type carrying out particular functions in the body.

Example bone and muscles. 1.5Marks

(b) Three animal tissue are Bones, muscles and Skin **1.5 marks**

Two plant tissues are Phloem and Xylem **1mark**

(c) **6 Marks@ 2Marks**

CELL WALL		CELL MEMBRANE
I.	Found in plants, Fungi, Algae and Bacteria	Found in animal cell and is second layer in plants
II.	It made up of chitin or cellulose	It made up of lipids and protein
III.	It allows materials such as gases, nutrients and water to pass into cell and waste to leave the cell	It is selective permeable membrane controlling movement of substance into and out of the cell

9. (a) Photosynthesis is the process by which green plants and other organisms use sunlight energy, carbon dioxide and water to manufacture their own food**. 2marks**

(b) Three importance of photosynthesis are

(i) Food production

(ii) Energy conservation

(iii) Balance of atmospheric gases

3 Marks@ 1mark

(c) $6\ CO_2 + 6\ H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6\ O_2$ **1mark**

Raw materials of photosynthesis are

- Carbon dioxide gas
- Water
- Sunlight **2 Marks**
- Chlorophyll

End products are

- Carbohydrates **2Marks**
- Oxygen gas

10. Tuberculosis; is the bacterial disease caused by bacteria known as Mycobacterium Tuberculosis. The disease mainly attach the lungs also may attack other organs apart from lungs.**2Marks**

    Mode of its transmission

    Tuberculosis is an air borne disease spread through contaminated air droplets

    Also can be transmitted through contaminated milk

    2marks

    Symptoms

    - Coughing blood
    - Night sweat **3marks**
    - Fever
    - Chest pain
    - Difficult breathing and loss of weight

      Prevention measures

      - **A**void close contact with infected person
      - Infected people should be treated as soon as they are diagnosed
      - Proper feeding and taking balance diet
      - Children should be immunized against bacteria causing tuberculosis
      - Mass education of tuberculosis
      - Avoid living in overcrowded and poorly ventilated places

        5 well explained points 2marks@=10marks

        Conclusion;Tuberculosis can be treared through the use of antibiotics such as **Stremptomycin, Rifampicin** and **Pyrazinamide** **3mark**

11. Biology Is the study of life. There are two branch of biology namely Botany means study of plant and Zoology means study of animal.

    It help us to understand our environment better, identifying different group of living things as well as to appreciate nature.**2marks**

    Biology is not an independent field of study but it is related to many fields such that is a foundation for many other scientific fields such as in

    Medicine

The study of biology has helped in finding the causes and ways in which disease spread. It also helped in finding method of preventing and eradicating diseases since Docters, Nurses, and mid wives have strong background in biology **3mark**s

Pharmacy

By studying biology the pharmacists understand the proper type of medicine for a particular diseases for treating human body and how it function hence dispense drugs to the patients in hospital, health centers and in pharmacies **3marks**

Agriculture

The knowledge of biology helps in improvements of crops. For example researchers have been able to produce crops which are resistance to diseases, mature faster and give high yields and then help in the prevention of diseases, production and improvement of animal products such as meat, milk and eggs **3marks**

Forestry

In studying biology we learn a lot about forests, on how to plant trees, the importance of planting trees and danger of cutting down trees since the forest help to preserve water sources, provide shade and shelter to livestock and wild life. **3marks**

Veterinary sciences

In studying biology gives foundation for understanding animal life process, behaviors and disease that attack them, their treatment and prevention since veterinary science is concerned with the health of animals and treating diseases that attack them **3marks**

Nutrition

Nutritionists and dieticians use the knowledge of biology to determines the kind of food suitable for people since nutrition is the study of how food is taken in and used in the body in relation to the body needs. It also explains various types of food substances and their significances to our body as well as the effects that can occur when there is lack of nutrients **3marks**